

OLARENWAJU ADISA

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RESEARCH PROFILE

Computer science lecturer and researcher in natural language processing, working on language technology for under-resourced languages and on the evaluation of online harms detection.

My doctoral direction concerns the safety of machine-translated health information in languages where no learned quality-estimation model exists and cannot easily be built. I am beginning work on the first evaluation resource for C'Lela, a Western Kainji language of north-west Nigeria that I have not found in FLORES, NLLB or the major multilingual corpora, and designing a reference-free check for clinically dangerous errors — altered dosages, substituted medicine names, dropped negations — that requires no trained model of the target language.

I also work on explainability and evaluation in abuse detection; my most recent paper reports a negative result on explanation-guided debiasing, tested against a frequency-matched random control. I supervise undergraduate research and have brought several student projects to peer-reviewed publication.

RESEARCH INTERESTS

Natural language processing; language technology for low-resource and under-represented languages; corpus and dataset construction; cross-lingual transfer and machine translation; machine translation evaluation and safety; detection of online abuse and harmful content; explainable AI and evaluation methodology; applied machine learning for health in African contexts.

EDUCATION

M.Sc. Computer Science, Afe Babalola University, Ado-Ekiti, Nigeria	2023 – 2025
CGPA 3.75 / 5.00. Thesis: Development of a Deep Learning Model for Heart Rate Estimation in Remote Photoplethysmography (rPPG). Awarded Best M.Sc. Research Proposal, Department of Computing.	
Postgraduate Diploma, Computer Science, Kebbi State University of Science and Technology	2020 – 2021
Aliero, Kebbi State, Nigeria. CGPA 4.03 / 5.00 (top 10% of cohort).	
Higher National Diploma, Computer Science, Rufus Giwa Polytechnic, Owo, Nigeria	2017 – 2019
Preceded by National Diploma in Computer Science, same institution (2014 – 2016).	

RESEARCH IN PROGRESS

Computational resources and safety evaluation for C'Lela

- Constructing the first machine-readable resources for C'Lela (~100,000 speakers, Zuru, Kebbi State): a structured lexicon derived from a 224-page SIL trilingual dictionary, and a health-domain parallel corpus developed with the Zuru speech community.
- Developing a reference-free clinical risk scorer that detects patient-safety-critical translation errors — altered dosages, dropped negations, substituted medicine names, fabricated content — without a learned quality-estimation model, since none exists at this resource level.
- Validating on the English–Hausa portion of AFRIDOC-MT through controlled error injection, benchmarked against BLEU, chrF, BERTScore and COMET.
- In correspondence with the AFRIDOC-MT team at Saarland University, Germany, regarding extension of their multilingual African health translation corpus to C'Lela.

PUBLICATIONS AND ACCEPTED PAPERS

Peer-reviewed — published

Adisa, O., Akinduyite, O., and Badeji-Ajisafe, B. (2024). A Comprehensive Review of Remote Photoplethysmography Techniques for Accurate and Robust Heart Rate Monitoring. 2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON), pp. 1–5. DOI: 10.1109/NIGERCON62786.2024.10927392.

Peer-reviewed — accepted

Adisa, O. and Ndubuisi, C. (2026). No Better Than Random: Evaluating LIME-Guided Saliency Penalisation for Fine-Grained Sexism Classification. Proceedings of the 10th Workshop on Online Abuse and Harms (WOAH 2026), co-located with EMNLP 2026, Budapest. Camera-ready September 2026.

Adisa, O., Bada, N. M., Tope-Oke, M. A., Abdul Raheem, B., and Akinduyite, O. C. (2026). An Explainable Machine Learning Framework for PCOS Screening in Low-Resource Healthcare Settings Using SHAP. African Symposium on Big Data, Analytics and Machine Intelligence (ASBAMI 2026); selected for a Springer journal special issue.

Akpofure, T. O., Olorunoje, T. A., Tope-Oke, M. A., Adisa, O., and Ayepeku, O. F. (2026). An Explainable AI Framework for Rare Variant–Microbiome Integration Across African Dietary Contexts. African Symposium on Big Data, Analytics and Machine Intelligence (ASBAMI 2026); selected for a Springer journal special issue.

Adisa, O., Okebule, T., and Arigibiola, M. (2026). Development of Deep Learning Based Object Detection Models for Cocoa Pod Disease Detection. Submitted to 2026 IEEE Nigeria International Conference on Digital Transformation (NIGERCON).

Olorunoje, T., Mumuni, A.-A., and Adisa, O. (2026). Development of a Cloud-Based Student Attendance System Using Face Recognition. Submitted to 2026 IEEE Nigeria International Conference on Digital Transformation (NIGERCON).

Under review / submitted

Adisa, O., Akinduyite, O., Badeji-Ajisafe, B., Obamiyi, E. S., and Orunnaiye, D. (2025). Deep Learning Framework for Non-Invasive Heart Rate Estimation Using Remote Photoplethysmography. Franklin Open (under review).

ACADEMIC EXPERIENCE

Assistant Lecturer, Department of Computing, Afe Babalola University

2025 – present

- Teach undergraduate courses in programming, computer architecture, digital design and systems analysis.
- Supervise final-year undergraduate research in natural language processing, computer vision and applied machine learning, several projects reaching peer-reviewed publication.
- Contribute to departmental research in applied artificial intelligence and language technology.

Graduate Researcher, Afe Babalola University

2023 – 2025

- Developed a non-contact heart-rate estimation framework using remote photoplethysmography under motion and illumination variability, combining vision transformers for spatio-temporal representation learning with recurrent models for temporal prediction.
- Evaluated performance on public benchmark datasets using MAE, RMSE and correlation metrics, with attention to cross-condition generalisation.

Lecturer (ICT), Kebbi State College of Nursing Science, Birnin Kebbi

2021 – 2022

- Delivered lectures in information and communication technology and supervised student projects.
- Direct exposure to clinical training and health communication practice in Kebbi State — the setting of current doctoral fieldwork.

RESEARCH SUPERVISION

Supervise final-year undergraduate research and mentor students through the full research cycle, from problem selection to peer-reviewed submission. Recent supervised projects:

- Fine-grained sexism classification and explanation-guided debiasing — accepted, WOA 2026 (EMNLP).
- Explainable machine learning for PCOS screening in low-resource healthcare settings — accepted, ASBAMI 2026 (Springer journal).
- Multi-omics integration with explainable AI — accepted, ASBAMI 2026 (Springer journal).
- Cocoa pod disease detection using object-detection architectures with mobile deployment — submitted, IEEE NIGERCON 2026.
- Cloud-based student attendance system using face recognition — submitted, IEEE NIGERCON 2026.
- Blockchain-based prototype for securing electronic health records.
- Student performance prediction using behavioural and demographic data.
- Fake news detection using natural language processing.

TEACHING

Undergraduate courses, Department of Computing, Afe Babalola University: Computer Programming; Object-Oriented Programming; Introduction to Digital Computer Design; Computer Architecture and Organisation; Problem Solving; System Analysis and Design.

Teaching materials are developed independently, with course content situated in Nigerian contexts to support student engagement.

TECHNICAL SKILLS

Languages: Python, C++, R, Rust, PHP, HTML, CSS.

Machine learning and NLP: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, NumPy, SciPy. Transformer fine-tuning (DeBERTa, BERT family), multi-seed experimental design, cross-lingual and transfer learning.

Machine translation evaluation: BLEU, chrF, TER, BERTScore, COMET; controlled error injection; inter-annotator reliability (Cohen's kappa, Krippendorff's alpha).

Explainability: LIME, SHAP, Integrated Gradients (Captum).

Tools: Git, LaTeX and Overleaf, Linux, Google Cloud Platform, Kaggle and Colab GPU environments.

AWARDS AND PROFESSIONAL MEMBERSHIP

- Best M.Sc. Research Proposal Award, Department of Computing, Afe Babalola University (2025).
- Student Member, IEEE Signal Processing Society.

CONFERENCE PARTICIPATION

- IEEE NIGERCON 2024 — oral presentation.
- ASBAMI 2026
- College of Sciences Research Day (COSRED), Afe Babalola University, 2025.
- Faculty Seminar, Kebbi State University of Science and Technology, 2023 — Energy-Efficient Computing.

REFEREES

Available on request.